

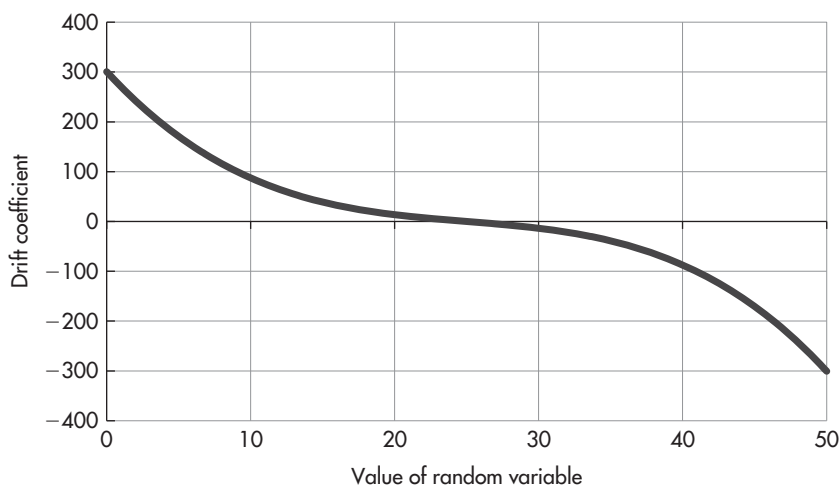


FIGURE 2.11 Target drift coefficient.

Source: Authors.

In this example, once we're done with the calculations in each of these 20 buckets, we'll have 20 points to plot on our diagnostic graph. The horizontal coordinate of each point will be the midpoint of the bucket, and the vertical coordinate of each point will be the average change of the observations in that bucket.

As an illustration, let's specify a nonlinear functional form for $f(x)$ and a constant value for the drift coefficient, $g(x)$. In particular, let's use the functional form illustrated in Figure 2.11.

Then let's simulate some data from the SDE we've specified and create our diagnostic graph using the method described above.

Figure 2.12 shows the resulting diagnostic graph. We see that this diagnostic graph is useful in that it helps identify a specification for $f(x)$ consistent with the mean reversion actually exhibited by the data.

The intention when creating this diagnostic graph is to identify the properties that are likely to be useful to the analyst trying to model the mean-reverting tendency in the data. The most straightforward way to do this is to select a functional form for $f(x)$ that has the ability to match the general shape traced out by the points in this diagnostic graph.

Of course, the same diagnostic graph can be created for the diffusion coefficient, $g(x)$. In this case, the standard deviation of the subsequent changes in each bucket would replace the average of the subsequent changes. The analyst then can specify a functional representation for $g(x)$ that most usefully captures the key properties of the diffusion coefficient, $g(x)$, as a function of x .

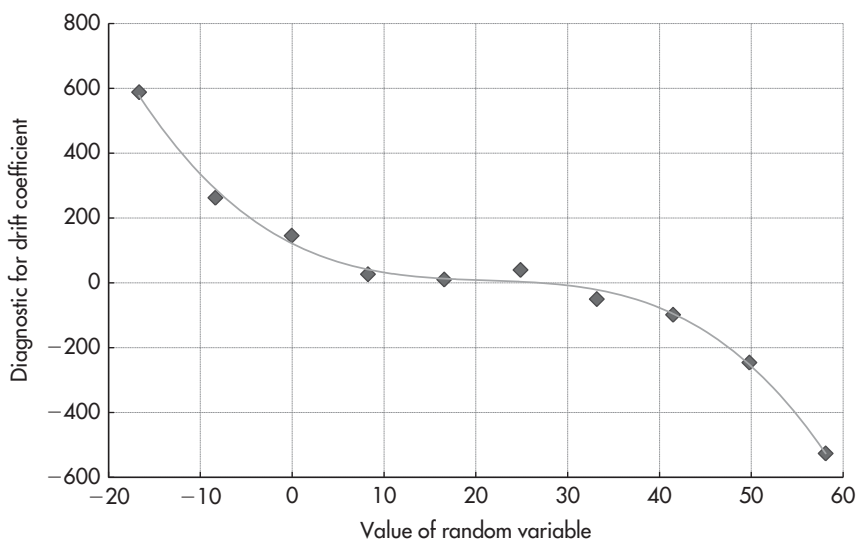


FIGURE 2.12 Diagnostic graph for drift coefficient.

Source: Authors.

As with the drift coefficient, $f(x)$, there are some technical conditions that must be satisfied for $g(x)$ in order to ensure that the SDE has a solution. In practice, we simply tend to make sure that $g(x)$ has some commonsense restrictions. For example, we'd like to restrict $g(x)$ to assuming only positive values, as a negative value for the diffusion coefficient would be equivalent to allowing time to run backward for our process.

In practice, a diagnostic graph for the diffusion coefficient often reveals relatively high levels of volatility at high distances from the mean (see Figure 2.21 for a real-world example). This corresponds to the fact that when the entry level into a trade is statistically attractive (far away from its mean), often also its volatility is higher than on average, and can be explained by the observation, that particularly good entry opportunities tend to occur in times of market turmoil (“buy when there is blood on the streets”). But this also means that the assumption of a constant volatility (i.e. independent of the distance from the mean), as in an OU process, can lead to underestimating the risk of a trade and to an overestimation of its Sharpe ratio. Using a diagnostic graph reveals whether the assumption of constant volatility is appropriate or needs to be corrected in order to reflect higher volatility and hence lower expected Sharpe ratios for a trade put on at extreme distance from its mean.

MODEL ESTIMATION

Once functional forms have been specified for the drift coefficient, $f(x)$, and for the diffusion coefficient, $g(x)$, the analyst must specify precise values for the functions $f(x)$ and $g(x)$ within the class of functions having the specified forms. For example, if the drift and diffusion coefficients have been specified with parametric representations, the analyst must choose specific, numeric values for these parameters. If nonparametric representations have been specified, the analyst still must assign specific values to the functions $f(x)$ and $g(x)$ over the relevant domain. For example, one could define the drift and diffusion for each bucket and run a Monte Carlo simulation by determining for each step of the simulation, in which bucket it lies, and using the values for this bucket to calculate the next step.

Most analysts will choose to specify numeric values for $f(x)$ and $g(x)$ to “most closely match” a set of historical data. For example, we can specify $f(x)$ and $g(x)$ so that the resulting SDE is the one from which the data are most likely to have been generated. This is the principle behind maximum likelihood estimation.

However, some analysts may prefer a different approach to specifying the drift and diffusion coefficients. For example, some analysts may want to specify $f(x)$ and $g(x)$ so that the moments of the resulting transition densities most closely match the empirical moments of the historical data. For example, some analysts will specify drift and diffusion coefficients so that their SDE has a conditional mean and a conditional variance that match the conditional means and variances of their historical data. More generally, we could match the theoretical and empirical expected values for a variety of functions of the data. Indeed, this is the general approach behind the method of moments. In some cases, using the method of moments will produce results that are precisely identical to the results obtained via maximum likelihood, though in general this need not be the case.

In some cases, an analyst may prefer to specify part or all of the SDE so that the theoretical moments of the process match some other data. For example, when the variable we’re modeling is the price of a security that serves as the underlying instrument for one or more derivative securities, the analyst may wish to specify the SDE so that the theoretical volatility of the process following the SDE is the same as the volatility implied by the prices of the derivative securities.

The important point to note is that different criteria can be used when specifying precise, numeric values for the drift coefficient, $f(x)$, and the diffusion coefficient, $g(x)$. The particular criteria used in any given application should be chosen in accordance with the goal of the analysis.

In some cases, an analyst may wish to conduct the analysis twice using different criteria in each instance, to see whether the results of the analysis

are robust to the choice of criteria. For example, if the ex ante Sharpe ratios of a trade are very different depending on whether the SDE is specified via maximum likelihood estimation or by matching the implied volatilities in the options markets, this difference is likely to be useful information for the analyst.

If an analyst elects to specify $f(x)$ and $g(x)$ via estimation using historical data, he needs to keep in mind that some estimation approaches can be problematic, conceptually and practically. For example, for some functional specifications of $f(x)$ and $g(x)$ (e.g. $f(x)$ is a line and $g(x)$ is a constant), the transition densities that comprise the likelihood function will have closed-form expressions. But for general choices of $f(x)$ and $g(x)$, there will be no closed-form representations of the likelihood function. In theory, the analyst could proceed in this case by calculating the likelihood function using numerical methods, such as simulations or by solving the partial differential equations for each transition density numerically, using finite difference grids. In practice, such methods are so numerically cumbersome as to be intractable (or at least impractical).

This problem also affects the method of moments in cases for which there are no closed-form representations of the moments of the process. When no closed-form expressions exist for the likelihood function, analysts sometimes approximate the likelihood functions. When the likelihood function is approximated, we refer to the approach as pseudo-maximum likelihood estimation (PMLE) or quasi-maximum likelihood estimation (QMLE).

In the case of PMLE, the approximations are based on the fact that the transition densities for the process are increasingly well approximated by normal densities when the time period between observations decreases. In other words, for small time intervals, the transition densities appear “locally normal.”

One problem with PMLE is that the estimators obtained by maximizing the pseudo likelihood function are generally not consistent, meaning the estimate can’t be guaranteed to converge to the true parameter value as more data become available. In addition, we typically don’t know the distribution of the estimator, even asymptotically, complicating the process of making inferences and testing hypotheses from the data.

At times, the lack of consistency and asymptotic normality will be an acceptable price to pay in order to work with an estimation procedure that is tractable in practice. In particular, this will be the case when the drift and diffusion are relatively “smooth” and particularly when the time between successive observations is small. Note that both of these conditions increase the extent to which the transition densities appear “locally normal.”

In our experience working with a large variety of financial variables over many years, the results obtained by using PMLE with daily data are virtually indistinguishable from the results one gets from using the exact likelihood

function. In other words, the cost of using PMLE rather than exact MLE tends to be small. Given that PMLE is much more tractable than most other methods in these cases, we recommend without any hesitation using PMLE when dealing with daily data.

While the scope of this book doesn't include the basics of maximum likelihood estimation (which we assume most analysts know), we can provide a sketch of the process.

Typically, a multivariate, nonlinear, numerical optimizer will choose parameter values so as to maximize the likelihood function, which is a function of both the historical data and the parameter values. The likelihood function is the product of the transition densities governing all the observations (i.e. the density of each observation conditional on the preceding observation). Clearly, each transition density is a function of two adjacent observations and of the historical data. From a purely computational perspective, working with the sum of functions often is easier than working with the product of functions. (For example, a product of many functions all of which are less than one may result in a value that is small relative to the precision of the machine performing the calculations.)

As it happens, the logarithm of a function achieves its maximum at the same point as the original function. And since the log of a product of functions is equivalent to the sum of the logarithms of the individual functions, this log likelihood function typically is very straightforward to calculate. As a result, it's often easier to maximize the logarithm of the likelihood function.

Once the log of the likelihood function has been specified, the analyst finds the set of parameter values that maximizes the value of this function, subject, possibly, to any constraints imposed by the analyst.

There are a variety of numerical algorithms for performing this nonlinear, numerical (possibly constrained) optimization. We won't discuss different algorithms in this book, except to note that gradient methods of one sort or another are almost always preferred by experienced analysts, as they tend to find local optima fairly quickly.

It's useful to note that these optima may be local rather than global, so it's good practice to start the optimization algorithm from a variety of starting points, to see whether the optimum obtained is robust to the choice of starting point. In our experience, most likelihood functions appear to be fairly well behaved, in the sense that we observe robust convergence to the global optimum. But this isn't always the case, and it's usually worth testing this proposition informally, to enhance our confidence in the results.

Once the drift and diffusion functions have been estimated, it's typically useful to plot the estimates, for a couple of reasons. First, plotting the coefficients can sometimes reveal nonsensical results, such as a diffusion coefficient that is negative over a portion of its range. Second, our intuition about the

behavior of a financial variable will be a function of the overall drift and diffusion functions rather than a particular parameter that appears in the drift or diffusion coefficient. Unless you're a savant, you're likely to gain intuition via graphical consideration of the coefficients that you wouldn't obtain by simple inspection of the parameter values.

In the theory of estimation, hypothesis testing is often an important topic. However, it's beyond the scope of this book. For our purposes, we'll simply mention that it's useful to have a sense for precision of our estimates of the drift and diffusion coefficients. As is often the case, a larger amount of data allows us to estimate the coefficients with greater precision than we can with a smaller amount of data. When it's important to gain some qualitative sense for the precision of our estimate, simulation often can be a useful exercise, for two reasons. First, in many cases, the asymptotic distributions of the estimators aren't known. And, second, even when we know the asymptotic distribution of an estimator, we don't know whether our particular data set is large enough for the asymptotic results to be relevant.

To simulate the distributions of the drift and diffusion coefficients, under the hypothesis that these coefficients have known values, we can simulate the process so as to generate a simulated sample equal in size to our actual sample. Then we can estimate the drift and diffusion coefficients using the simulated sample as our historical observations. If we repeat this procedure, say, 10,000 times, we'll obtain 10,000 individual estimates of the drift and diffusion coefficients, which we could use to quantify the estimation error of our estimators or for the purpose of making inferences and/or testing hypotheses, under the hypotheses that the data were generated from the SDE used to generate our simulated data.

As it happens, this approach to quantifying estimator error, making inferences, and/or testing hypotheses tends to produce better results than relying upon the asymptotic results in cases where they're known, since in this case we don't rely on the assumption that our data are of sufficient length for the results to approach an asymptotic limit.

CALCULATING CONDITIONAL EXPECTATIONS AND PROBABILITY DENSITIES

In the previous section, on estimation, we discussed the notion that transition densities tend to appear locally normal as the time between successive observations decreases, even for rather general diffusion processes. And we used this result to motivate our discussion of PMLE as an approach toward estimation of the drift coefficient, $f(x)$, and the diffusion coefficient, $g(x)$. But when we're assessing the merits of a trade over a horizon of more than, say,

one week, we can rely less on the local normality of our transition densities. In these cases, we need to be able to calculate transition densities and their properties using other approaches.

As we mentioned previously, it is possible to solve for transition densities by solving the underlying partial differential equations that they must satisfy. One approach to doing this is to use finite difference grids. While this approach is straightforward at a conceptual level, it tends to be cumbersome in practice.

A more useful approach is to compute the transition densities and their moments numerically via stochastic simulations. The basic idea is simple. For any SDE, we simply simulate paths between the desired starting time and starting point until the desired ending time. By simulating more paths, we can compute these densities and their moments with as much accuracy as we'd like.

If we'd like to know the mean of a transition density, we can simulate sample paths and calculate the sample average of the terminal values. If we'd like to know the standard deviation of a transition density, we can simulate sample paths and calculate the sample standard deviation of the terminal values.

Similarly, if we'd like to consider the transition density in its entirety, we can simulate sample paths and produce a histogram of the resulting terminal values. An example of this is given in Figure 2.13.

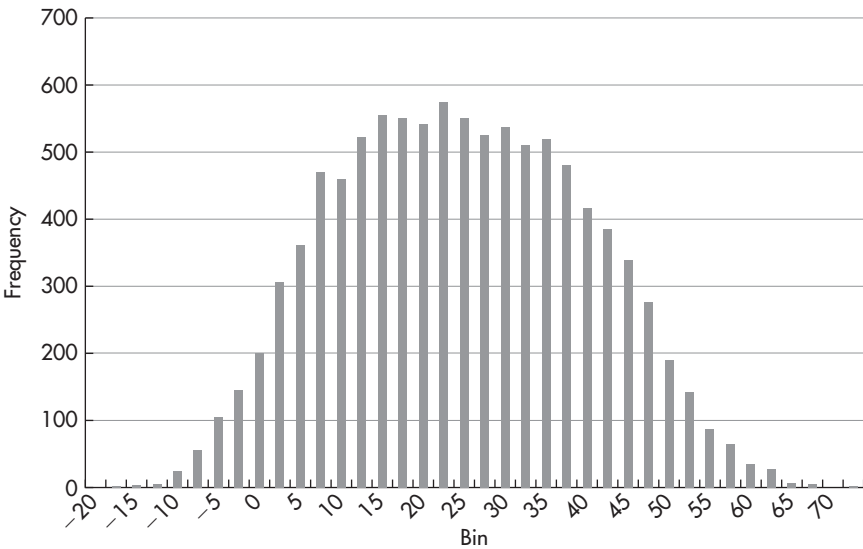


FIGURE 2.13 Histogram of simulated values.
Source: Authors.

One way to increase the smoothness of the depicted transition density is to simulate an increasing number of sample paths. However, another way is to impose some smoothness conditions on the results directly. For example, we can produce a smooth, continuous, and differentiable function representing a transition density nonparametrically by applying a kernel density to the data.

For example, one form of kernel density is the Gaussian kernel density, which uses the Gaussian function (i.e. a normal density) as its kernel. The functional form of the Gaussian kernel density, Q , is given by:

$$Q(x; h) = \sum_{i=1}^N K_i(x; h)$$

$$K_i(x; h) = \frac{1}{h\sqrt{2\pi}} e^{\frac{-(x-y_i)^2}{2h^2}}$$

where x is the argument of the kernel density, Q ; K is our Gaussian kernel; N is the number of data points, y_i , used to estimate the nonparametric density; and h is the bandwidth parameter, which determines the smoothness of the Gaussian kernel density.

An analogy to a histogram might be helpful. Recall that a histogram sums the number of observations that fall into a particular bin or bucket. If an observation falls into a bucket, it increases the frequency of that bucket by one, and it doesn't change the frequency of any other buckets.

In contrast, when evaluating the kernel density for any particular argument, every data point makes some contribution to the value of the kernel density evaluated at that argument. If a data point is close to the argument, it makes a relatively large contribution, and if the data point is far from the argument, it makes a relatively small contribution.

The relative sizes of the contributions of each data point are determined in part by the bandwidth parameter, h , which acts like the standard deviation of the Gaussian kernel. When h is small, the standard deviation of each Gaussian kernel is small. In that case, data points that are close to the argument make particularly large contributions to the Gaussian kernel, and data points that are far from the argument make particularly small contributions to the Gaussian kernel, since the standard deviation causes the Gaussian density to be especially peaked around its mean. When h is large, the standard deviation of the Gaussian kernel is large, and there is relatively less difference between the contribution to the kernel density made by data points that are close to the argument and those that are far from the argument. In this sense, the bandwidth parameter, h , plays a key role in determining the tradeoff between the smoothness and the granularity in the Gaussian kernel density.

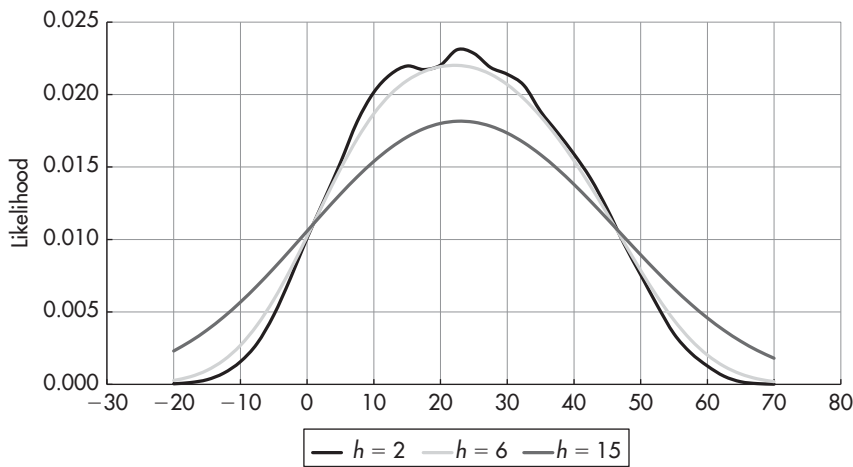


FIGURE 2.14 Nonparametric kernel densities with different bandwidths.
Source: Authors.

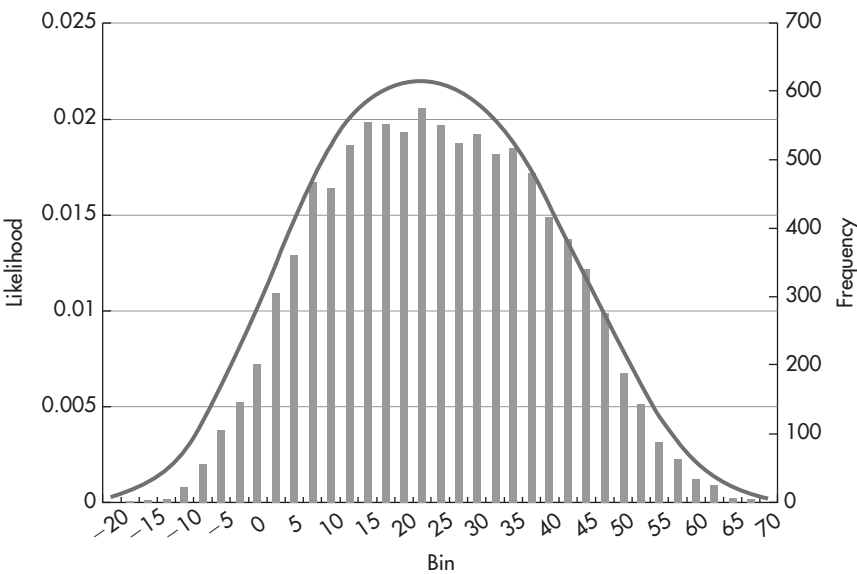


FIGURE 2.15 Nonparametric kernel density and histogram.
Source: Authors.

The bandwidth parameter, h , is specified by the analyst. There are a number of optimal criteria that have been developed for this bandwidth parameter under various assumptions. But in our experience, it's fine simply to use the "eyeball metric" in which the choice is made to produce a nonparametric density that appears most useful from simple visual inspection. To illustrate, Figure 2.14 shows nonparametric densities corresponding to the simulated data of Figure 2.13 for three different choices for the bandwidth parameter: $h = 2$, $h = 6$, and $h = 15$.

In our view, $h = 2$ produces a density that is insufficiently smooth, while $h = 15$ produces a kernel density that may have been smoothed too much.

Figure 2.15 shows the kernel density corresponding to bandwidth parameter $h = 6$, along with the histogram from Figure 2.13.

In our view, this choice for the bandwidth parameter produces the nonparametric density with the most useful tradeoff between smoothness and granularity.

CALCULATING CONDITIONAL, EX ANTE RISK-ADJUSTED RETURNS

Sharpe Ratio as One Measure of Risk-Adjusted Return

Traders and investors tend to use a wide variety of measures to quantify risk, including value at risk, probability of threshold loss, and expected maximum drawdown. Each of these has merit, and we recommend viewing risk from multiple perspectives whenever possible. So we encourage using all these measures, and others, to gain an understanding of risks.

But for the purpose of this chapter, we'll focus our discussion of risk on a single measure: standard deviation. In choosing this risk measure, we don't mean to imply that standard deviation is somehow superior to other measures of risk. But it is relatively simple to calculate, and most people have at least a rudimentary intuition about standard deviations, whereas this isn't always the case when dealing with other risk measures.

One of the fundamental results of financial economics is that investors face tradeoffs between the risks to which they expose their portfolios and the returns that they reasonably can expect to earn. As a result, in any ex ante evaluation of a trade or portfolio, or in any ex post performance attribution exercise, one should consider risks and returns simultaneously.

There are a variety of ways in which analysts can compare risks and returns. The one we focus on in this chapter is to calculate risk-adjusted returns, that is we adjust returns for the risks incurred in the pursuit of those returns.